

Scalable Deep Learning for Stress and Affect Detection on Resource-Constrained Devices

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Abstract—Psychological stress in human beings has been on a meteoric rise over the last few years. Chronic stress can have fatal consequences such as heart disease, cancer, suicide and so on. It is thus imperative to detect stress early on to prevent health risks. In this work, we discuss efficient and accurate stress and affect detection using scalable Deep Learning methods, that can be used to monitor stress real-time on resource-constrained devices such as low-cost wearables. By making inferences on-device, we solve the issues of high latency and lack of privacy which are prevalent in cloud-based computation. Using the concept of Early Stopping - Multiple Instance Learning, we build specialized models for stress and affect detection for 3 popular datasets in the domain, that have very low inference times but high accuracy. We introduce a metric η_{comp} to measure the computational savings from the use of these models. On average, our models show an absolute increase of 10% in overall accuracy over the benchmarks, computational savings of 95.39%, and an 18x reduction in inference times on a Raspberry Pi 3 Model B. This allows for efficient and accurate real-time monitoring of stress on low-cost resource-constrained devices.

Index Terms—stress, affect detection, physiological signals, resource-constrained, on-device, scalable, time-series, deep learning

I. INTRODUCTION

Stress is a psychological response to mental, emotional, or physical challenges that humans encounter. When an individual is overwhelmed with stress, the body goes into *fight-or-flight* mode, which triggers the adrenal glands to release hormones that cause an increase in heart rate and blood pressure. According to a 2018 study [1], 74% of the people surveyed stated that they were unable to cope with stress. Learning to identify when an individual is stressed and what is causing the stress can greatly improve one's mental and physical health. According to the American Psychological Association, chronic stress is linked to six leading causes of death - heart disease, cancer, lung ailments, accidents, cirrhosis of the liver and suicide. There is also evidence that stress plays a crucial role in the development of cardiovascular diseases, diabetes and asthma [2]–[4]. It is therefore necessary to predict chronic stress at the early stages to prevent various health risks.

This calls for an efficient way to track the real-time emotional state of human beings.

Affect is a psychological concept that describes human emotions or feelings. Affect recognition gives rise to a wide array of opportunities to monitor a human being. Russell [5] proposed that three independent dimensions - Arousal, Dominance and Valence - are necessary and sufficient to characterize any emotional state. Affect recognition may be used to recognize other emotions such as amusement or anger, and thus may help a doctor to maintain a complete record of a patient's emotional state. This highly valuable data can be used to observe patterns with respect to the emotional behavior of an individual to provide correct and effective treatment.

To prevent permanent damage from the effects of stress, there is great need to develop a method for stress detection that can accurately, continuously and unobtrusively monitor stress on a real-time basis, without affecting the lifestyle of the person.

Physiological data analysis is a natural method for stress and affect detection. In other affect detection methods based on facial expressions or speech, the actual emotion can be suppressed relatively easily [6]. However, it is not possible to easily control the natural responses of the nervous system. This makes physiological signals possibly the truest and best indicators of stress and emotion.

Deep Learning architectures can be designed for stress and affect detection [7], as these algorithms can learn high-level features from raw physiological data. This eliminates the need for domain expertise and hardcore feature extraction. But Deep Learning requires high computational power, contrary to simple Machine Learning methods. A cloud server is thus required to process complex computations. There is however a significant round trip time associated with cloud computation. A real-time stress monitoring system would be impacted adversely by the latency. Other issues such as privacy also cause great concern. Health data of individuals is extremely private, and there is high risk of this data being compromised when a cloud server is used for processing.

No better solution exists to these concerns than the deployment of scalable deep learning models on resource-constrained devices. On-device processing of physiological data can provide real-time stress and affect monitoring, while also keeping the individual's data private as it never leaves the device.

In this paper, we discuss a scalable Deep Learning approach for efficient and accurate stress and affect detection on a resource-constrained device. The rest of this paper is organized into 7 sections. Section II talks about the recent developments in deploying Deep Learning models on resource-constrained devices. Section III provides an overview of the datasets considered while Section IV provides extensive data analysis. The core concepts used and the models built are discussed in Sections V & VI. Our results are presented in Section VII, from which we draw conclusions in Section VIII.

II. BACKGROUND

Deployment of efficient Machine Learning and Deep Learning models on resource-constrained devices has been an active area of research for the past decade. Kargupta et al. [8] provide early insights towards efficient data analysis on light-weight computing devices. On resource-constrained devices, reducing model size and minimizing inference time are paramount. The former can be done by compressing deep learning models using Pruning, Trained Quantization and Huffman Coding [9], [10], while the latter necessitates the designing of light-weight models.

Wu et al. [11] proposed Quantized CNN, an efficient framework to simultaneously speed-up computation and reduce the memory overhead of CNN models. ProtoNN is a novel machine learning algorithm introduced in [12], inspired by K-Nearest Neighbors (KNN) algorithm. It addresses the problem of accurate real-time prediction on resource-constrained devices. Kumar et al. [13] introduced Bonsai, a tree-based algorithm, for efficient prediction on IoT devices. Bonsai was able to deliver predictions in the order of milliseconds on resource-constrained devices. It also achieves as much as 30% better accuracy compared to state-of-the-art techniques for resource-efficient Machine Learning. However, these models have not been designed specifically for handling sophisticated data such as sequential data.

Recent developments have brought about new techniques to handle sequential data. Dennis et al. [14] propose a concept, where they couple *Multiple Instance Learning* with *Early Stopping* technique to formulate EMI (Early Stopping - Multiple Instance Learning). EMI can be used to build deep learning models that make predictions with low inference times for sequential data-points, while maintaining state-of-the-art accuracies.

As mentioned earlier, physiological signals are known to be great indicators of stress and emotion. These signals can be extracted from low-cost wearable devices such as the Emotiv EPOC and SHIMMER. At present, research on Machine Learning and Deep Learning for stress and affect detection has been focused on improving accuracy. However,

the models proposed are computationally expensive and cannot be used to perform on-device predictions on resource-constrained devices. Thus, connection to a cloud server capable of performing such demanding computations is required. The consequences of this are issues arising from latency, low bandwidth, energy consumption and privacy concerns. Making predictions on-device would allow for seamless and continuous real-time monitoring of stress without any cloud connectivity.

In our work, we consider 3 datasets from the domain of stress and affect detection and deploy specialized Deep Learning models on a resource-constrained device, using the concept of EMI. The research work showcases superior accuracy with drastically lesser inference times.

III. DATASETS

We considered three popular datasets from the domain of stress and affect recognition - WESAD, SWELL-KW and DREAMER - for our experiments. Table I summarizes the descriptions of these datasets.

A. WESAD

WESAD [15] is a multimodal dataset for stress and affect detection. Various physiological parameters were measured from 15 participants, using two devices - a chest-worn device and a wrist-worn device. The participants were put through a study protocol, through which the aim was to elicit 3 affective states - Neutral, Stress and Amusement. The participants marked their affective states in a self-evaluation sheet after each phase in the protocol. The various modalities measured were Blood Volume Pulse (BVP), Electrocardiogram (ECG), Electrodermal Activity (EDA), Electromyogram (EMG), Respiration (RESP) and Temperature (TEMP). In addition, Accelerometer (ACC) readings were also recorded. In [7], the Artificial Neural Network architecture discussed was able to achieve a mean accuracy of 87.18% in predicting the affective states, using the modalities extracted from the chest-worn sensor.

B. SWELL-KW

The SWELL-KW dataset [16] was created for the purpose of research on stress and user modeling. In this dataset, emotions were elicited in 25 participants by making them perform tasks such as writing reports, making presentations, reading emails, and searching for information. The working conditions of the participants were manipulated using stressors such as time-pressure and email interruptions. The various modalities recorded in the experiment were Computer Interaction, Facial Expressions, Body Posture and Physiological sensor values. The Physiological modalities recorded were Heart Rate Variability (HRV) and Skin Conductance (SC). Based on the participants' responses to a questionnaire, 3 conditions - Neutral (N), Interruption (I) and Time-Pressure (T) - were analyzed. In [17], an accuracy of about 64.10% was reported on the N vs. I&T binary classification task.

TABLE I
DATASET DESCRIPTIONS

Dataset	Number of Participants	Physiological Modalities	Affective States	Elicitation Methods	Devices Used	Miscellaneous
WESAD	15	BVP ECG EDA EMG RESP TEMP ACC	Neutral, Stress, Amusement	Video Clips, Trier Social Stress Test	RespiBAN Professional, Empatica E4	700 Hz ECG, EMG, EDA, RESP, TEMP, and ACC (Chest) 64 Hz BVP, 4 Hz EDA, 4 Hz TEMP and 32 Hz ACC (Wrist)
SWELL-KW	25	HRV SC	Neutral, Interruption, Time-Pressure	Report Writing, Email	TMSi Mobi	2-channel 2048 Hz HRV, 1-channel 2048 Hz SC
DREAMER	23	EEG ECG	Arousal, Dominance, Valence	Film Clips	Emotiv EPOC, SHIMMER	14-channel 128Hz EEG, 2-channel 256 Hz ECG

C. DREAMER

DREAMER [18] is a dataset aimed at Emotion Recognition using low cost devices. DREAMER consists of Electroencephalogram (EEG) and Electrocardiogram (ECG) readings recorded through contact sensors from 25 participants. The participants were asked to watch 18 film clips that elicited various emotions, following which they were asked to provide an emotional assessment. The participants marked their scores on a 5-point scale for Arousal, Dominance and Valence. In [18], the authors estimated Arousal, Valence and Dominance states with accuracies of 61.84%, 62.32% and 61.84% respectively.

IV. DATA ANALYSIS

Physiological modalities have commonly been used in Machine Learning methods for stress detection [19]. These modalities are extracted from wearable sensors over a period of time, and are thus time-series data. Efficient and accurate methods are required to estimate stress from such data, on resource-constrained devices.

A. Data Visualizations and Observations

Extensive exploratory data analysis showed that each affective state had a distinct *class signature*. These were values that were highly representative of a particular affective state.

Fig. 1 and 2 show the values of Electrodermal Activity (EDA) for the Neutral (N) and Interruption & Time-Pressure (I&T) affective states respectively. There is a visible difference in the EDA values for these states - while the values for the N state lie between 0 and 0.02, the values for the I&T states range from 0 to just over 0.10. However, quite a few values labeled I&T lie between 0 and 0.02, and do not truly represent the affective state.

The Inductive Respiratory Plethysmograph (RESP) values for the Neutral and Stress affective states of the WESAD dataset are shown in Fig. 3 and 4. The RESP values lie between -5 and 5 for the Neutral state, while in the Stress state, the values are often higher than 10 and close to 15. We may thus consider these peaks to be highly representative of the Stress state. As was the case in the SWELL-KW dataset,

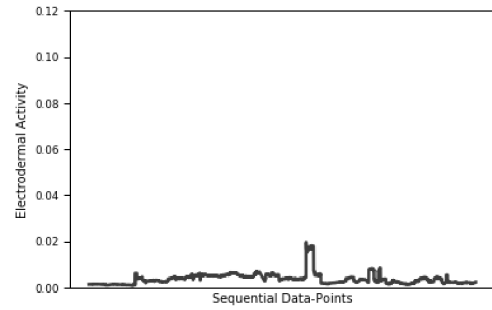


Fig. 1. Neutral Affective State, SWELL-KW Dataset

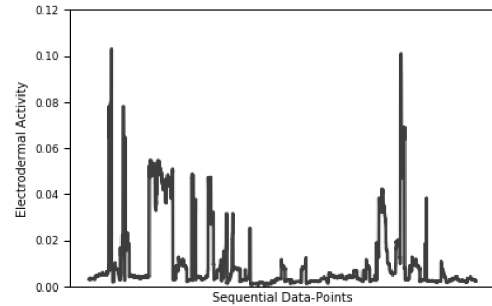


Fig. 2. Interruption and Time-Pressure Affective States, SWELL-KW Dataset

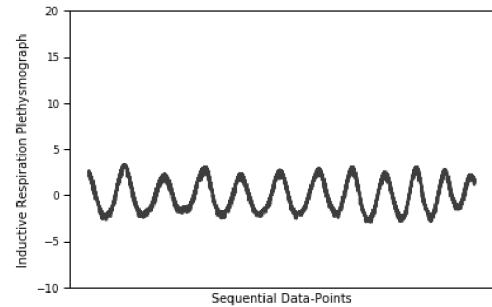


Fig. 3. Neutral Affective State, WESAD Dataset

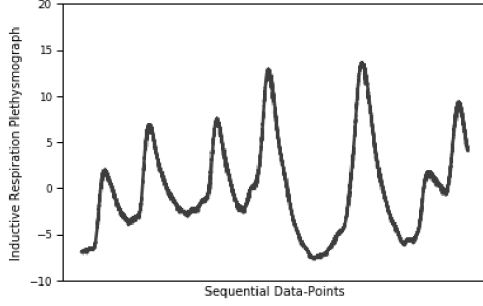


Fig. 4. Stress Affective State, WESAD Dataset

the values that make up the class signature for Stress are not the most frequent, and many values lie between -5 and 5 which are characteristic of the Neutral state.

B. Inferences

We infer from our observations that the class signature is quite infrequent, and does not form the majority of values to which the state has been assigned as label. Thus, sophisticated methods are required to correctly identify the class signature amid all the uncharacteristic data, which may be considered as noise. Using the concept of EMI (Section V), we aim to capture the class signature with minimum computation and maximum accuracy on resource-constrained devices.

V. EARLY STOPPING - MULTIPLE INSTANCE LEARNING

As shown in Section IV, the class signature does not form a majority of the values to which the class has been assigned as label, while there is a significant amount of noise present in the data. Existing approaches based on RNNs analyze the entire sequential data-point and thus fail in capturing the class signature and predicting the class label efficiently. This problem can be alleviated by creating finer labels, thus increasing the granularity from a *bag* level label to a *sub-instance* level label, as discussed below.

A. Multiple Instance Learning

Fig. 5 illustrates the concept of Multiple Instance Learning. $Z_{i,0}$ and $Z_{i,1}$ are sub-instances which are present in the bag χ_i .

In Multiple Instance Learning, the dataset, which is made up of sequential data-points, is split into mutually exclusive and exhaustive bags. The sequential data-points within a bag are divided into multiple overlapping sub-instances of equal width ω . That is, a sub-instance consists of ω sequential data-points. Sub-instance width ω must be chosen such that it is closer to the width of the label's class signature.

The concept of Multiple Instance Learning holds true irrespective of the base architecture and is further explained using an RNN.

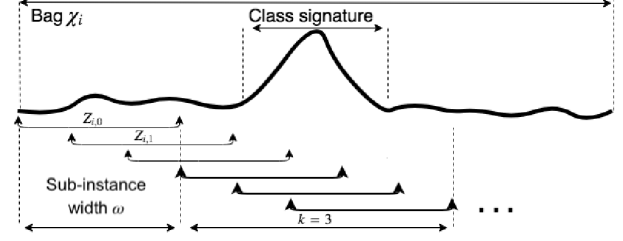


Fig. 5. Multiple Instance Learning

B. Working of MI-RNN

According to [14], a bag can be represented as $\chi_i = [Z_{i,1}, \dots, Z_{i,T-\omega+1}]$, where $Z_{i,\tau} = [X_{i,\tau}, \dots, X_{i,\tau+\omega-1}]$. Here, T represents the number of sequential data-points in a bag, $Z_{i,\tau}$ denotes the τ^{th} sub-instance and X represents a data-point. Initially, the label of the entire bag is set to the class label since we do not have the information regarding the potential noise present in the bag. However, only a few of the sub-instances contain the class signature and hence represent the class label, while the other points in the bag represent noise and should ideally be labeled as -1. This way, the correct sub-instances are captured for training.

We choose a parameter k to represent the number of consecutive sub-instances in each bag that contain the class signature. The other sub-instances may be considered as noise. So the classifier is trained to identify the class based on the k sub-instances chosen per bag. The LSTM further re-estimates the correct sub-instances in each training step.

C. Analysis of MI-RNN

We consider the entire dataset to be made up of positive and negative bags. A few sub-instances in the positive bags contain the class signature, whereas negative bags do not contain any such sub-instances. Let us consider the simple problem of binary classification for the purpose of this analysis. The label vector for negative bags contains -1 for every sub-instance whereas in the case of positive bags, the labels may be 1 or -1. We initially assume all sub-instances of positive bags to be labeled 1.

The aim of training is to obtain a classifier that can correctly classify each sub-instance present in the bags. The algorithm below is used to obtain an optimal solution by re-estimating the classifier and the set of correct sub-instances in every iteration.

Consider $\mathbf{w}$ as the classifier to be learned, χ_i^P as the i^{th} positive bag and χ_i^N as the i^{th} negative bag. $Z_{i,\tau}^P$ and $Z_{i,\tau}^N$ denote the τ^{th} sub-instances from the i^{th} positive and negative bags respectively. We consider k to be the number of consecutive sub-instances in each bag that contain the class signature. S is the set of start indices of the k correct sub-instances in each bag. According to [14], we have,

$$\mathbf{w}^0 = \arg \min_{\mathbf{w}} \sum_{i,\tau} (1 - \mathbf{w}^T Z_{i,\tau}^P)^2 + \sum_{i,\tau} (-1 - \mathbf{w}^T Z_{i,\tau}^N)^2 \quad (1)$$

$$S_{r+1} = \bigcup_{i=1}^n \bigcup_{\tau=k_1}^{k_1+k-1} (i, \tau),$$

$$\text{where } k_1 = \arg \max_{k_1} \sum_{\tau=k_1}^{k_1+k-1} \langle \mathbf{w}^r, Z_{i,\tau}^P \rangle, \forall i, \forall r \geq 0 \quad (2)$$

$$\mathbf{w}^{r+1} = \arg \min_{\mathbf{w}} \sum_{(i,\tau) \in S_{r+1}} (1 - \langle \mathbf{w}, Z_{i,\tau}^P \rangle)^2 + \frac{1}{T} \sum_{i,\tau} (-1 - \langle \mathbf{w}, Z_{i,\tau}^N \rangle)^2, \forall r \geq 0 \quad (3)$$

Thus, we estimate a classifier $\mathbf{w}$ and the correct sub-instances to learn from in every iteration, until convergence. The above algorithm for training converges at a linear rate to the global optima.

D. Early Stopping (E-RNN)

The biggest drawback of an LSTM is that during prediction, it iterates through all data-points before making a prediction. This is computationally very expensive and has an adverse impact on inference times on resource-constrained devices. As many of the data-points are noise and only a few data-points truly represent the class label, the concept of Early Stopping is brought into the picture. The output of every step of the LSTM is piped to a classifier, rather than just the last step. The network is trained to minimize the sum of loss of the classifier's output at each time step, as shown in Equation 4,

$$\min \sum_i \sum_{t=1}^T (l(w^T o_{i,t})) \quad (4)$$

where w is the weight of the fully connected layer, and $o_{i,t}$ is the output of the t^{th} step of the LSTM when processing the data-point X_i .

Therefore, if the classifier predicts a label with a probability greater than the threshold probability $\hat{p}$ (confidence level), the LSTM delivers the prediction without iterating through the rest of the steps. It is efficient to predict the probability after every k steps. This greatly reduces the time of inference on a resource-constrained device.

E. EMI-RNN

The EMI-RNN integrates the concept of MI-RNN and E-RNN. While MI-RNN chooses the best sub-instances to learn from, E-RNN delivers early prediction. Interestingly, the techniques are more effective when put together rather than when trained separately [14]. Fig. 6 represents the joint EMI-RNN architecture. The data are converted into sub-instances as explained in Section V-B and the LSTM loss function is replaced by the loss function given in Equation 4.

VI. METHODOLOGY

In this section, we discuss the application of EMI models to the datasets we considered. Due to the inherent differences in the data, different EMI parameters have to be chosen for each dataset. Hence, optimal bag sizes and sub-instance widths are chosen for each dataset.

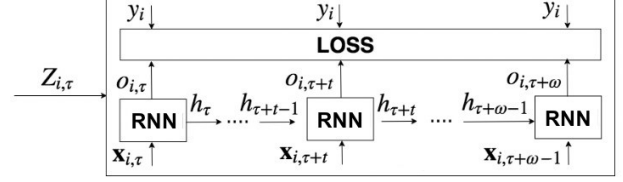


Fig. 6. EMI-RNN Architecture

A. Dataset Pre-Processing

For our experiments on the WESAD dataset, we considered the modalities from the chest-worn device as they were superior compared to those from the wrist-worn device [15]. In total, 8 features were selected to be passed as input to our models - ECG, EDA, EMG, RESP, TEMP and ACC (x, y and z values).

We considered only the Physiological modalities (HRV and SC) to build our models for the SWELL-KW dataset. The data, which was initially sampled at 2048 Hz, was downsampled to 60 Hz. A total of 3 features were thus selected to be sent as input to the model - 2-channel HRV and 1-channel SC values.

For the DREAMER dataset, we downsampled the ECG data from 256 Hz to 128 Hz. This was done in order to use both ECG and EEG features as inputs to the model. Totally 16 features were used - 14-channel EEG and 2-channel ECG values.

B. Bag Size

After pre-processing, the data-points in each dataset had to be split into bags. For each dataset, we arrived at an ideal bag size based on our analysis, keeping in mind the minimization of inference time and hence computation required. We also wanted to maximize the number of bags which contained the class signature, to improve the performance of our models. The bag sizes for each dataset can be found in Table II.

The sampling frequency was a decisive factor in calculating the bag size. The ideal bag size for each dataset was either equal to or a fraction of the sampling frequency (after pre-processing). This is represented in the equations below,

$$B = \left\{ b : \frac{\nu_s}{m} \leq b \leq \frac{\nu_s}{n}, b \in \mathbb{N} \right\} \quad (5)$$

$$T = \arg \max_b f(b), b \in B \quad (6)$$

where,

B is the set of possible bag sizes,

ν_s is the sampling frequency,

m and n are parameters,

$f(b)$ is the model's accuracy when bag size is b ,

T is the ideal number of sequential data-points in a bag, which is the bag size.

Based on our analysis, and keeping in mind the aspect of real-time application of the models, we determined the values of m and n to be 4 and 1 respectively.

In Fig. 7, 3 bags with the Stress label from the SWELL-KW dataset are shown, separated by dotted lines. The size of each bag is 20 sequential data-points.

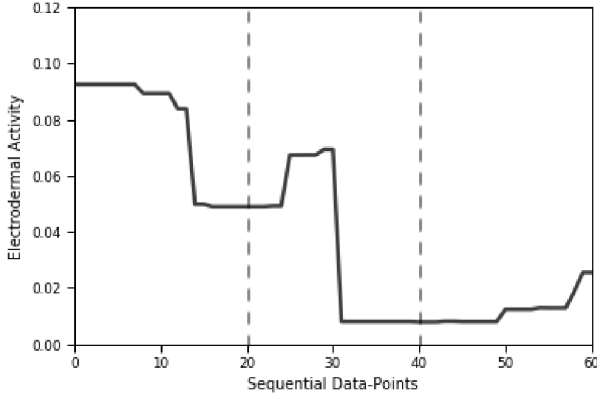


Fig. 7. Bags Labeled as Stress, SWELL-KW Dataset

C. Sub-instance Width

The sub-instance width for each dataset was chosen on closer inspection of the bags themselves. While a large sub-instance width would mean lesser computation per bag, it would also make our models less sophisticated due to increased noise in the sub-instances. So we chose a sub-instance width which could, on average, just encapsulate the class signature in each bag. Table II shows the sub-instance width for each dataset. The equations below represent the method of obtaining the ideal sub-instance width.

$$W = \left\{ \omega_s : \frac{T}{p} \leq \omega_s \leq \frac{T}{q}, \omega_s \in \mathbb{N} \right\} \quad (7)$$

$$\omega = \arg \max_{\omega_s} g(\omega_s), \omega_s \in W \quad (8)$$

where,

W is the set of possible sub-instance widths,

T is the bag size,

p and q are parameters,

$g(\omega_s)$ is the model's accuracy when ω_s is the sub-instance width,

ω is the ideal sub-instance width.

From our analysis, p and q were derived to be 3 and 2 respectively. That is, ideal sub-instance width was between one-third and half the bag size.

Fig. 8 shows that it is possible to capture the class signature with a sub-instance width of about 8 sequential data-points.

D. Model Specifications

We discuss three different models to which the concepts of Multiple Instance Learning and Early Stopping are applied viz., EMI-LSTM, EMI-GRU and EMI-FastGRNN [14], [20]. LSTMs and GRUs are popular RNN architectures that are known to perform well on Sequential Data. In [20], the authors

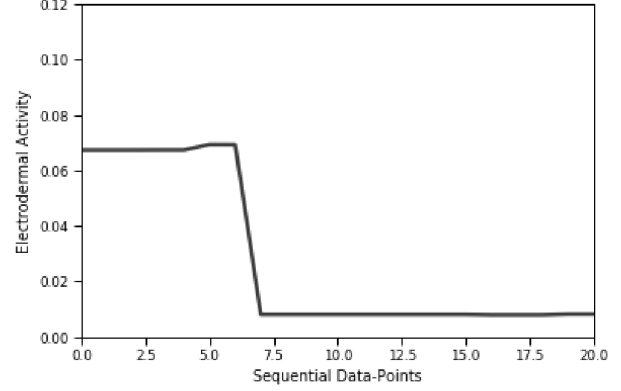


Fig. 8. Single Bag Labeled as Stress, SWELL-KW Dataset

introduced the FastRNN and FastGRNN model architectures, which employ residual connections [21], [22] to outperform unitary RNNs with respect to training time, prediction cost and accuracy. The prediction cost of FastGRNN is often an order of magnitude below other model architectures, despite nearly matching the accuracies. FastGRNN utilizes low-rank, sparse and quantized representations to learn a classifier of very low model size, but near state-of-the-art accuracy and very low inference times.

The parameters considered for each of the models are mentioned in Table II. These models are benchmarked against the aforementioned datasets and the results are compared and contrasted with the state-of-the-art techniques in Section VII.

TABLE II
EMI PARAMETERS

Dataset	Bag Size	Sub-instance Width (w)	k
WESAD	175	88	2
SWELL-KW	20	10	3
DREAMER	128	48	2

VII. RESULTS AND OBSERVATIONS

All the experiments were carried out on a Raspberry Pi 3 Model B and the results were recorded.

A. Benchmark Models

To prove the superiority of EMI models for the datasets considered, the results have to be compared with the benchmark model for each dataset. The benchmark model for each dataset was deployed on a Raspberry Pi 3 Model B. The results were recorded and compared with those of our models.

For WESAD, an ANN was implemented [7] taking the chest-based modalities as features. For SWELL-KW, an SVM with RBF kernel was implemented, with the physiological modalities as features [17]. In the case of DREAMER, according to [18], separate SVMs with RBF kernel were trained with EEG and ECG modalities as features, for prediction of Arousal, Dominance and Valence.

B. Metrics

In addition to accuracy and inference time, the metrics mentioned below were used to judge the performance of the models. The inference times reported are for prediction on 1 bag.

1) Computational Savings:

$$\eta_{\text{comp}} = \frac{I_B - I_M}{I_B} \times 100 \quad (9)$$

where,

I_B represents the inference time of the benchmark model for prediction on 1 bag,

I_M represents the inference time of the EMI model considered for prediction on 1 bag.

Computational Savings (η_{comp}) is indicative of the computation saved based on reduction in inference time of the model, relative to that of the benchmark. It is represented as a percentage value.

2) Early Savings:

$$\eta_{\text{ES}} = \left(1 - \frac{1}{n} \sum_{i=1}^n \frac{T_{\text{pred}_i}}{T} \right) \times 100 \quad (10)$$

where,

n represents the number of bags,

T_{pred_i} represents the number of data-points traversed to make a prediction for bag χ_i ,

T represents the total number of data-points in a bag.

Early Savings (η_{ES}) is a metric indicative of the computation saved due to Early Stopping. As a consequence of Early Stopping, not all data-points in a bag need to be traversed to make a prediction. Early Savings is the fraction of data-points that have not been traversed, over the total number of data-points in a bag, represented as a percentage.

C. WESAD Results

The EMI-LSTM, EMI-GRU and EMI-FastGRNN were evaluated on the WESAD dataset as mentioned in Section VI-D. The results are summarized and presented in Table III. The EMI-FastGRNN is observed to perform the best with 97.55% accuracy and an inference time of 12.8 ms. The inference time is approximately 13 times lesser than the state-of-the-art method and this translates to a 92% computational saving. An early saving of 96.07% was obtained.

TABLE III
WESAD RESULTS

Model	Accuracy (%)	Inference Time	η_{comp} (%)	η_{ES} (%)
Benchmark	87.18	160 ms	-	-
EMI-LSTM	93.09	55.2 ms	65.50	87.55
EMI-GRU	85.54	44.9 ms	71.94	82.62
EMI-FastGRNN	99.86	12.8 ms	91.99	96.07

D. SWELL-KW Results

For the SWELL-KW dataset, all the 3 models considered outperform the benchmark model as observed in Table IV. EMI-FastGRNN is the best performing model with an accuracy of 87.87% and an extremely low inference time of 3 ms. This is 35 times lower than the benchmark model, which is observed to provide an inference time of 104 ms. This translates to a very high computational saving of 97.1%. While the EMI-FastGRNN is observed to provide an early saving of 29.55%, the EMI-GRU provides an early saving of 66.5%.

TABLE IV
SWELL-KW RESULTS

Model	Accuracy (%)	Inference Time	η_{comp} (%)	η_{ES} (%)
Benchmark	64.09%	104 ms	-	-
EMI-LSTM	86.79%	12.0 ms	88.46	20.87
EMI-GRU	84.74%	9.8 ms	90.60	66.50
EMI-FastGRNN	87.87%	3.0 ms	97.10	29.55

E. DREAMER Results

We created separate models for the prediction of Arousal, Dominance and Valence, for the DREAMER dataset. The performance of EMI-LSTM and EMI-GRU are close, with the EMI-LSTM having the edge as far as accuracy is concerned. Table V showcases the performance of the models in comparison to the benchmark model. The inference time of the EMI-LSTM is approximately 8.5 times lesser when compared with the benchmark. This translates to a computational saving of 87.88%. A maximum early saving of 67.5% was obtained on the EMI-GRU.

TABLE V
DREAMER RESULTS

Model	Accuracy (%)			Inference Time	η_{comp} (%)	η_{ES} (%)
	Arousal	Dominance	Valence			
Benchmark	62.32	61.84	62.49	406 ms	-	-
EMI-LSTM	67.75	68.87	65.60	49.2 ms	87.88	20.52
EMI-GRU	64.69	67.02	63.09	38.0 ms	90.64	67.50
EMI-FastGRNN	54.29	52.95	44.62	11.8 ms	97.09	2.21

VIII. CONCLUSION

The concept of Early Stopping - Multiple Instance Learning was applied to 3 datasets from the domain of Stress and Affect Detection. The EMI models outperformed the state-of-the-art models with respect to accuracy and inference time. On average, there was an absolute increase of 10% in overall accuracy, while the inference times were observed to be 18 times lesser when compared to the state-of-the-art models. The average computational savings across the 3 datasets was 95.39%. The EMI-FastGRNN outperformed the EMI-LSTM and EMI-GRU for the WESAD and SWELL-KW datasets.

The EMI models thus facilitate accurate stress and affect detection, with exceptional accuracy. Due to the decreased complexity and low inference times of these models, they can be pushed onto resource-constrained devices to make real-time on-device predictions.

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